
Healthcare Throughput & Patient Flow Analysis

Case Study & Operational Insights Report

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1. Project Overview

This project analyzes patient throughput, wait times, satisfaction, and departmental performance using a real-world healthcare operations dataset. The goal is to identify bottlenecks, quantify their impact on patient experience, and provide actionable recommendations to improve service delivery.

The analysis integrates **Python (EDA, feature engineering, visualization)** and **Power BI (interactive dashboard)** to create a comprehensive operational intelligence solution.

2. Business Problem

Healthcare organizations face increasing pressure to improve patient flow, reduce wait times, and maintain high satisfaction. Long delays create operational strain, reduce patient trust, and increase the likelihood of escalation or adverse outcomes.

This project addresses three core questions:

1. How do patient arrivals and throughput vary over time?
2. Which departments and hours contribute most to wait time delays?
3. How do wait times influence patient satisfaction and admission patterns?

3. Dataset Summary

The dataset includes **9,216 patient encounters** from April 2023 to October 2024, with fields covering:

- Admission date & time
- Department referral
- Wait time
- Satisfaction score
- Admission flag
- Age, gender, race

Feature engineering expanded the dataset with:

- Hour, day, month, year
- Age groups
- Wait time categories
- Admission likelihood indicators

All Power BI visuals were built using a **cleaned CSV exported directly from the Python workflow**, ensuring consistency and reproducibility across the analytical pipeline.

4. Methods & Tools

Python (Jupyter Notebook)

- Data cleaning
- Feature engineering
- Exploratory data analysis
- Statistical correlation
- Visualizations (Matplotlib & Seaborn)

Power BI Dashboard

- KPI summaries
- Interactive filters
- Multi-page operational and satisfaction analysis
- Department × hour heatmaps
- Trend analysis

Together, these tools create a unified analytical workflow.

5. Key Findings

A. Patient Volume & Throughput

- Daily patient volume fluctuates between **8 and 30 patients**, with clear peaks.
- Hourly arrivals surge between **10 AM and 2 PM**, creating predictable congestion.
- These peak windows align with elevated wait times and reduced satisfaction.

B. Wait Time Performance

- Average wait time: **35.3 minutes**
- Median wait time: **35.0 minutes**
- **5,467 patients** waited longer than 30 minutes.
- Departments such as **Neurology, Physiotherapy, and Gastroenterology** show consistently high wait times (30–40 minutes).
- Heatmaps reveal red zones (45–50+ minutes) during late morning and early afternoon.

C. Admission Patterns

- Overall admission rate: **50%**
- Highest admission departments: **Renal (53%)**, Neurology & Orthopedics (50%)
- Lowest admission department: **General Practice (48%)**

D. Satisfaction Insights

- Only **12.5%** of patients report high satisfaction.
- Satisfaction scores cluster heavily at **4/10**, indicating systemic dissatisfaction.
- Satisfaction declines sharply as wait times exceed **30 minutes**.
- Age groups show similar satisfaction levels, suggesting system-wide issues rather than demographic-specific concerns.
- Hourly satisfaction dips during peak congestion hours.

E. Correlation Analysis

- Strong negative correlation between wait time and satisfaction.
- Admission likelihood varies with age and department.
- Patient load correlates with increased wait times.

6. Operational Recommendations

A. Improve Peak-Hour Throughput

- Increase triage staffing between **10 AM–2 PM**
- Implement fast-track pathways for low-acuity patients
- Rebalance departmental load during peak windows

B. Target High-Wait Departments

- Conduct workflow audits for Neurology, Physiotherapy, and Gastroenterology
- Evaluate scheduling practices and resource allocation
- Introduce queue management or digital check-in systems

C. Enhance Patient Experience

- Improve communication during delays
- Provide real-time wait time updates
- Train staff on expectation-setting and service recovery

D. Strengthen Data Governance

- Investigate “Unassigned” department entries
- Standardize referral and department coding practices

7. Conclusion

This project demonstrates how combining Python-based analytics with Power BI visualization can uncover meaningful insights into patient flow, operational bottlenecks, and satisfaction drivers. The findings highlight clear opportunities to improve throughput efficiency, reduce wait times, and enhance patient experience.

By addressing peak-hour congestion, optimizing departmental workflows, and improving communication, healthcare organizations can significantly improve both operational performance and patient satisfaction.

This case study showcases the ability to integrate data engineering, statistical analysis, and business intelligence into a cohesive, actionable healthcare operations solution.